

Lecture 5: The Inductor

the other ideal energy-storage element — and a summary of the three

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■ flux linkage ■ current ■ voltage ■ charge ■ power, energy

The other way to store energy

Lecture 4: capacitor

charge on plates, *electric* field

$$i = C dv/dt$$



Today: inductor

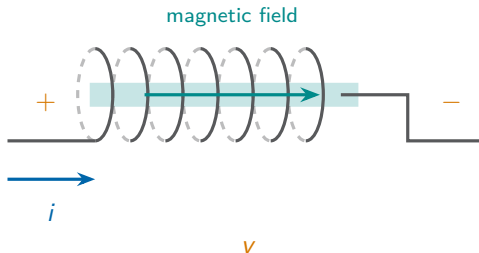
current in a coil, *magnetic* field

$$v = L di/dt$$

The same equation with v and i **exchanged**.

That is no coincidence. Try to guess each result today before it appears.

A coil, and what it does

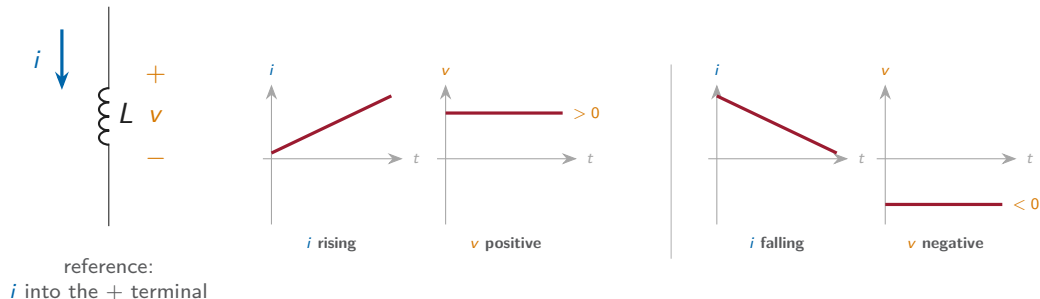


A current in the coil sets up a magnetic field running through it.

The interesting part is what happens when you try to **change** that current.

The **coil** is the object on the bench. The **inductor** is the circuit element we model it with — from here on, **coil** for the thing, **inductor** for the element.

The back EMF



The voltage always **opposes the change** that produced it — the **back EMF**.
The marks never move: with i into the + terminal, a rising current makes v **positive** and a falling one makes it **negative**.
The faster the change, the bigger it is: $v \propto di/dt$.

Inductance: how hard it pushes back

$$v = L \frac{di}{dt} \qquad 1 \text{ H} = 1 \frac{\text{V} \cdot \text{s}}{\text{A}}$$

One henry gives one volt of back EMF when the current rises at one ampere per second.

L is larger with

more **turns** — each one catches the field again
a different **shape** — long and thin, or short and fat
and above all what is **inside** — iron instead of air
multiplies L by hundreds

A 1000-turn air coil,
1 cm², 5 cm long:

$$L \approx 2.5 \text{ mH}$$

nH to a few mH is normal.
Henries mean an iron core.

Volt-seconds: the inductor's version of charge

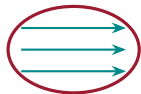
$$\varphi(t) = \varphi(t_0) + \int_{t_0}^t v(\tau) d\tau \quad \Longleftrightarrow \quad v = \frac{d\varphi}{dt}$$

	capacitor (Lecture 4)	inductor (today)
accumulate which variable?	the current i	the voltage v
the running total	$q = \int i d\tau$	$\varphi = \int v d\tau$
measured in	amp-seconds = coulombs	volt-seconds = webers
so	$i = dq/dt$	$v = d\varphi/dt$

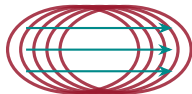
Everything in this lecture follows from that one definition.

Why it is called *flux linkage*

φ is the magnetic field inside the inductor, counted once for every turn it passes through.



few turns: small φ



many turns: large φ

More turns, a fatter winding, or iron instead of air $\Rightarrow$ more field $\Rightarrow$ larger φ .

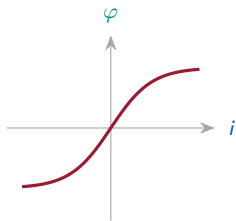
The **same three things** that made L large.

And a *changing* φ is the back EMF. That is Faraday's law.

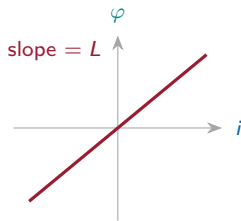
q is countable stuff; volt-seconds are not. Nothing piles up — but φ behaves exactly as charge did.

What an inductor is

A two-terminal element is an **inductor** if its (i, φ) pairs lie on a **curve in the i - φ plane**.



in general



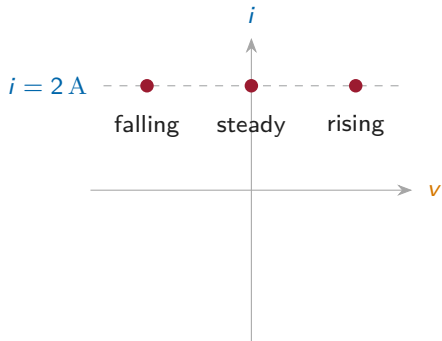
LTI: a straight line

The bent curve is a real iron core:
past some current it **saturates**.

In this course every inductor is
linear and time-invariant.

And again, note what has *not* happened

An inductor has **no characteristic** in the v - i plane either.



No f , no g :

$$v(t) \neq f(i(t)), \quad i(t) \neq g(v(t))$$

One current, **any** voltage — it follows how i is *changing*, not what it is.

The LTI inductor

$\varphi = L i$ — a straight line whose slope is the **same** L we met three slides ago.
Differentiate:

$$v = L \frac{di}{dt}$$

differential

$$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v(\tau) d\tau$$

integral

The voltage is whatever the **rate of change** of current demands.
The current is the **running total** of everything the voltage has done.

Four things that follow

Short at DC	Current is continuous	It remembers	It holds energy
i constant $\Rightarrow \frac{di}{dt} = 0 \Rightarrow v = 0.$	Flip a switch, step the source: i is the <i>same</i> just after as just before. It always takes time to change.	$i(t)$ depends on the <i>whole history</i> of v .	$w = \frac{1}{2}Li^2$, set by the present current.

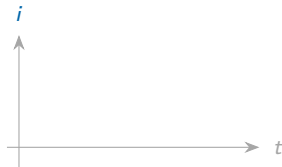
To a DC source, an inductor is just a piece of wire.

Example: hold a constant voltage across it

$V = 5\text{ V}$ across $L = 10\text{ mH}$, from $i(0) = 0$



voltage held



Example: hold a constant voltage across it

$$V = 5 \text{ V across } L = 10 \text{ mH, from } i(0) = 0 \quad i(t) = \frac{V}{L} t = 500 t \text{ [A]}$$



voltage held



current ramps

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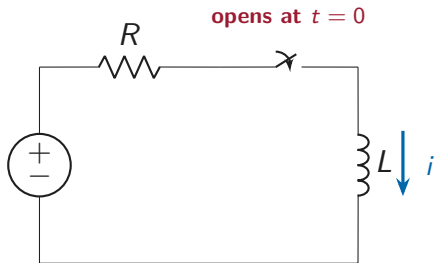
current ramps

The inductor is **integrating** the voltage.

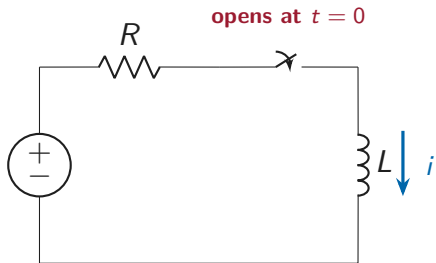
5 A after 10 ms — Example 4.1 with the variables swapped.

What if you try to stop the current?

A steady $i = I_0$ is flowing.



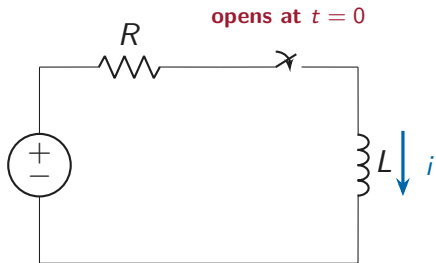
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The switch opens: i must reach zero at once.

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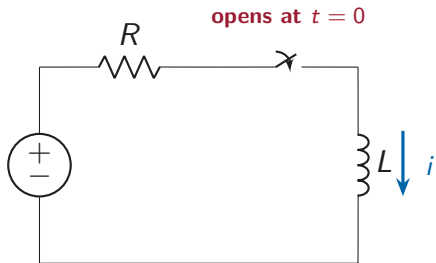


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$$\frac{di}{dt} \rightarrow -\infty, \text{ so } v = L \frac{di}{dt} \rightarrow -\infty.$$

What if you try to stop the current?



A steady $i = I_0$ is flowing.

The switch opens: i must reach **zero** at **once**.

$$\frac{di}{dt} \rightarrow -\infty, \text{ so } v = L \frac{di}{dt} \rightarrow -\infty.$$

The inductor will do **whatever it takes** to keep the current going — so the voltage climbs until the **air breaks down**, and $\frac{1}{2}LI_0^2$ crosses the gap as a spark.

Where does the energy go?

Raise i from zero. The inductor pushes **back**, so the circuit does **work**.

$$p = v i = L i \frac{di}{dt}$$

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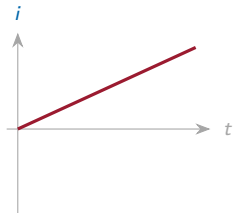
Raise i from zero. The inductor pushes **back**, so the circuit does **work**.

$$p = v i = L i \frac{di}{dt} \quad \Rightarrow \quad \int_0^I L i di$$

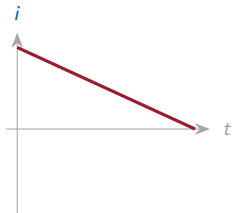
$$w = \frac{1}{2} L i^2 = \frac{\varphi^2}{2L}$$

No resistance anywhere — so none of it became heat.

Now run it backwards



i rising: $p > 0$



i falling: $p < 0$

$di/dt < 0$, so v changes sign: the inductor stops resisting the current and starts **driving** it.

It hands back exactly what it was given.

$w \geq 0$ always $\Rightarrow$ **passive**. Nothing lost either way $\Rightarrow$ **lossless**.

Example: opening that switch was this handover done all at once — as a spark.

So where was it kept?

capacitor

work done *separating* charge
kept in the **electric** field

inductor

work done *building up* current
kept in the **magnetic** field

Not in the wire, and not in the moving charge — in the **field** around it.

A field holds energy without consuming any.

That is the whole reason these two are lossless and a resistor is not.

For the mechanically minded

$$v = L \frac{di}{dt} \quad \longleftrightarrow \quad u = \frac{1}{k} \frac{dF}{dt}$$

circuit		mechanical
current i	$\leftrightarrow$	force F
voltage v	$\leftrightarrow$	velocity u
capacitance C	$\leftrightarrow$	mass m (Lecture 4)
inductance L	$\leftrightarrow$	spring compliance $1/k$
$\frac{1}{2}Li^2$	$\leftrightarrow$	$F^2/2k$

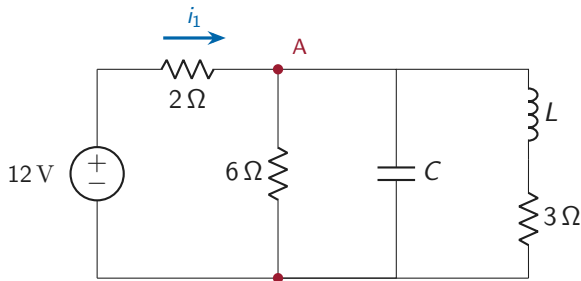
A stiff spring is a small inductance.

The three LTI elements, side by side

	LTI resistor	LTI capacitor	LTI inductor
characteristic in	$v-i$	$v-q$	$i-\varphi$
that characteristic	$v = Ri$	$q = Cv$	$\varphi = Li$
terminal equation	$v = Ri$	$i = C dv/dt$	$v = L di/dt$
unit	ohm, Ω	farad, F	henry, H
memory	none	the past of i	the past of v
at DC it is	itself	an <i>open</i> circuit	a <i>short</i> circuit
cannot jump	—	its voltage	its current
energy w	—	$\frac{1}{2}Cv^2$	$\frac{1}{2}Li^2$
verdict	passive, <i>lossy</i>	passive, <i>lossless</i>	passive, <i>lossless</i>

Swap $v \leftrightarrow i$, $q \leftrightarrow \varphi$, $C \leftrightarrow L$ and the last two columns **turn into each other**.

A circuit with all three

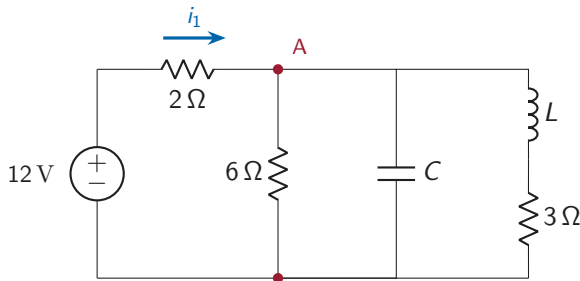


$$C = 100\ \mu\text{F},\ L = 50\ \text{mH}.$$

Nothing has changed for a long time.

Every branch voltage and current = ?

A circuit with all three



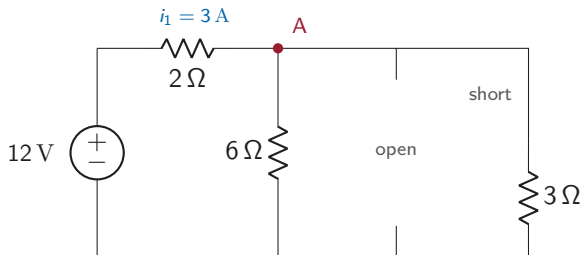
$$C = 100 \mu\text{F}, \quad L = 50 \text{ mH}.$$

Nothing has changed for a long time.

Every derivative is zero, so by the table: C is an **open circuit**, L is a **short circuit**.

What is left is Lecture 2's resistive circuit.

The steady state, solved



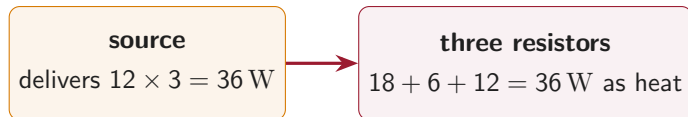
$$i_1 = \frac{12}{2 + (6 \parallel 3)} = 3\text{ A}$$

$$v_A = 12 - 2(3) = 6\text{ V}$$

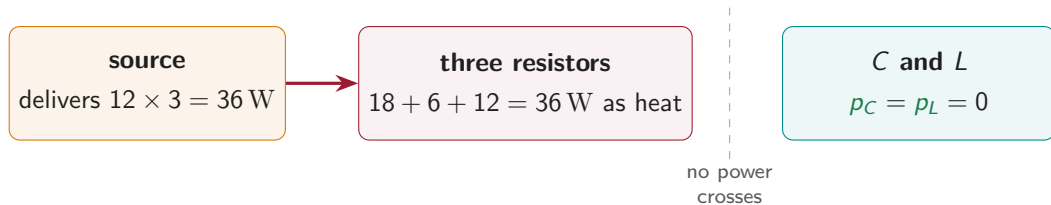
$$i_{6\Omega} = 1\text{ A}, \quad i_L = i_{3\Omega} = 2\text{ A}, \quad v_C = 6\text{ V}, \quad v_L = 0, \quad i_C = 0.$$

$$\text{Stored: } w_C = \frac{1}{2}(10^{-4})(6)^2 = 1.8\text{ mJ}, \quad w_L = \frac{1}{2}(0.05)(2)^2 = 0.1\text{ J}.$$

Where is the power going?



Where is the power going?



$$p_C = v_C i_C = 6 \times 0 = 0, \quad p_L = v_L i_L = 0 \times 2 = 0.$$

They **hold** energy. They do not consume any.

Summary

- An **inductor** fights changes in current. Integrate the voltage: $\varphi = \int v d\tau$ in **volt-seconds** (webers), as charge was amp-seconds. An **inductor** is a curve in the i - φ plane — not a resistor.
- **LTI inductor**: $v = L di/dt$, or the integral form. Short circuit at DC; current cannot jump; it remembers.
- **Inductance** L in henries — set by the turn count, the winding's shape and what is inside it. One henry is large.
- Starting the current means **work against** that opposing voltage: $w = \frac{1}{2} Li^2$, kept in the **magnetic field** and handed back in full. Forcing the current to jump demands a voltage impulse — in a real switch, a spark.
- In a **DC steady state**: C open, L short, and neither absorbs power.

Next lecture

Dependent sources

Sources whose value is set by a voltage or a current *somewhere else* in the circuit — how a transistor or an op-amp gets into a circuit diagram.

Questions?

Notes for this lecture: see the course repository.